Certified Robustness from One Setpoint Step: A Safe Acceptance Test for Model-Free PID Tuning

Pavel Otta[1], Daniel Pachner[1], Jiří Dostál[1] and Vladimír Havlena[2]

***Abstract*— Model-free PID tuners improve a loop from closed-loop data, but none guarantees its robustness, measured by the peak M_s of the sensitivity function. We show that the setpoint step these tuners already apply is enough to certify it. In a stable digital loop, the increments of the error after a unit setpoint step are the impulse response of the sensitivity function. One finite record therefore gives a lower and an upper bound on M_s, under one assumption on the unrecorded rest of the response. The same record also certifies the stability and the M_s of any other controller before it is applied. With these two certificates, any iterative tuner can accept only tunings with $M_s \le M^*$, and never apply a tuning with $M_s > M^*$. We add them to SPIN, a tuner that counts the turns of the error in phase portraits, in two steps: certified SPIN checks each tuning after applying it, safe SPIN before. On 19 simulated loops with $M^* = 1.7$, basic SPIN ended above M^* on up to 9 loops and applied tunings above M^* in 57% of its iterations; certified SPIN ended below M^* on all loops, but still applied such tunings in 45%; safe SPIN never applied one.**

I. Introduction

Let C be the controller, P the plant, $L = CP$ the loop transfer function, $S = 1/(1+L)$ the sensitivity function and $T = 1 - S$ the complementary sensitivity function. The peak $M_s = \max_\omega |S|$ measures robustness: $1/M_s$ is the shortest distance from the Nyquist curve of L to the critical point -1, so M_s near 1.4 is safe and near 2 is poor [1]. Model-based tuning rules reach a chosen M_s [1], [2]. Model-free tuners adjust the gains from closed-loop data instead, by iterative feedback tuning [3], extremum seeking [4], virtual reference feedback tuning [5], setpoint overshoot [6] or step-response inspection [7]. They target performance or a response shape; none guarantees the M_s of its result, or of the tunings it tries on the way. Once a loop runs, its M_s is rarely checked at all [8], [9].

The $\mathcal{H}_\infty$ norm of a system, and hence M_s, can be estimated from data: by power iterations over repeated dedicated experiments [10], [11], by data-driven estimation [12], from limit cycles induced in closed loop [13], or from impulse-response data under a decay assumption, with a one-sided guarantee [14]; margins can be bounded from data [15]. These methods need dedicated experiments, give estimates or one-sided bounds, and are not built into a tuner. Safe Bayesian optimization tunes controllers without leaving a safe set [16], with probabilistic guarantees under a Gaussian-process model of the cost.

[1]University Centre for Energy Efficient Buildings (UCEEB), Czech Technical University in Prague, Buštěhrad, Czech Republic. Corresponding author: pavel.otta@cvut.cz

[2]Department of Control Engineering, Faculty of Electrical Engineering, Czech Technical University in Prague, Czech Republic.

TABLE I
Main symbols

C, P, $L = CP$	controller, plant, loop transfer function
$S = \frac{1}{1+L}$, $T = 1 - S$	sensitivity, complementary sensitivity
$M_s = \max \lvert S \rvert$, M^*	robustness measure, its target
e_n, $a_n = e_n - e_{n-1}$	error after a unit setpoint step, its increments
$\hat{S}(\theta)$	estimate of S from the record $e_0, \dots, e_N$
τ	bound on the unrecorded tail $\sum_{n>N} \lvert a_n \rvert$
$\underline{M}$, $\overline{M}$	certified lower and upper bound on M_s
$\rho = C'/C - 1$	relative change from C to the next controller C'
Γ_k, N_k, $\bar{N}_k$	SPIN portraits, their turn counts, limits

The idea. A tuner already applies a setpoint step in every iteration. After a unit step, the increments $a_n = e_n - e_{n-1}$ of the error e are the impulse response of S (Lemma 1), so their Fourier sum is S itself. One record thus shows $|S|$ at every frequency, and with it the peak M_s (Fig. 1). We turn this elementary fact into guarantees, for a chosen target M^*:

1) *bounds*: one finite record gives a lower and an upper bound on M_s that hold with certainty, not estimates (Section II, Theorem 1);
2) *tests for any iterative tuner*: a test that passes only tunings with $M_s \le M^*$ (Theorem 2), and a test of the next tuning *before* it is applied (Theorem 3), which keeps every applied tuning below M^* (Section III);
3) *application to SPIN* [7]: two variants built on the tests, certified SPIN, which ends below M^*, and safe SPIN, which never applies a tuning above M^* (Section IV, Corollaries 1 and 2), and why its turn counts alone cannot certify M_s (Lemma 2, Proposition 2).

Section V confirms the results in simulation. Table I lists the main symbols.

II. A Certificate from One Setpoint Step

A digital controller runs with the sampling period T_s. Seen at the sampling instants, the loop it closes is linear and time invariant, with transfer functions $C(z)$, $P(z)$, L, S and T as above. On the unit circle, $z = \mathrm{e}^{j\theta}$ with the normalized frequency $\theta = \omega T_s \in [0, \pi]$, and $M_s = \max_\theta |S(\mathrm{e}^{j\theta})|$; we write $M_s(C)$ when the controller matters. All these quantities refer to the digital loop, the loop that actually runs; e (upright) is Euler's number.

Lemma 1 (The step contains S): Let the loop be stable and have one degree of freedom (the controller acts on the error $e = r - y$, with setpoint r and output y), and let L be strictly proper (the measurement precedes the control action). Let e_n be the error after a unit setpoint step at $n = 0$, applied from rest and without disturbances, so that $e_{-1} = 0$ and

$e_0 = 1$, and let $a_n = e_n - e_{n-1}$. Then

$$S(\mathrm{e}^{j\theta}) = \sum_{n\geq 0} a_n \, \mathrm{e}^{-jn\theta}. \tag{1}$$

Proof: Let $R(z)$ and $E(z)$ be the z-transforms of the setpoint and the error. A unit step has $R = 1/(1-z^{-1})$, and $E = SR$, so $S = (1 - z^{-1})E$, whose coefficients are the increments a_n. In a stable loop, a_n decays exponentially, so the series converges on the unit circle. □

A real record $e_0, \ldots, e_N$ is finite, so one assumption covers the increments it does not show.

Assumption 1 (Long enough record): The loop is stable, and the record $e_0, \ldots, e_N$ satisfies $\sum_{n>N} |a_n| \leq \tau$ for a known τ.

This is the only assumption the certificate needs. A valid τ shrinks geometrically as the record gets longer (Section V checks it with a four times longer record). No finite record can prove that the loop is stable or that the tail is small: an unstable or slow mode may appear only after the record ends. So Theorem 1 bounds M_s of a loop assumed stable, and Theorem 3 later certifies the stability of each next controller from the record of the current one.

By (1), the record gives the estimate $\hat{S}(\theta) = \sum_{n=0}^{N} a_n \mathrm{e}^{-jn\theta}$ of S at every frequency. Its largest value on a frequency grid differs from M_s for three reasons, which Theorem 1 bounds one by one: the unrecorded tail (term τ), the gaps between grid points (term $hJ/2$), and high frequencies, where a grid would have to be very fine (term $B/\sin(\bar{\theta}/2)$).

Theorem 1 (Certified bounds from one record): Let Lemma 1 and Assumption 1 hold. Let G be a uniform grid on $[0, \bar{\theta}]$, $0 < \bar{\theta} \leq \pi$, with spacing h and containing both ends. For any integer n_c, let

$$J = \textstyle\sum_{n=0}^{N} |n - n_c| \, |a_n|,$$
$$B = \textstyle\sum_{n=1}^{N-1} |a_{n+1} - a_n| + |a_N| + 2\tau.$$

Then $\underline{M} \leq M_s \leq \overline{M}$, where

$$\underline{M} = \max_{\theta\in G} |\hat{S}(\theta)| - \tau,$$
$$\overline{M} = \max\Big\{ \max_{\theta\in G} |\hat{S}(\theta)| + \tfrac{hJ}{2} + \tau, \; 1 + \frac{B}{\sin(\bar{\theta}/2)} \Big\}, \tag{2}$$

and the second term is omitted if $\bar{\theta} = \pi$. If τ equals the true remaining variation $\sum_{n>N} |a_n|$, then both bounds converge to M_s as $N \to \infty$ and $h \to 0$ with $\bar{\theta} = \pi$.

Proof: Tail. By (1), $S - \hat{S} = \sum_{n>N} a_n \mathrm{e}^{-jn\theta}$, so $|S - \hat{S}| \leq \tau$ at every θ. This gives $\underline{M}$ and the term τ in $\overline{M}$. *Grid.* On $[0, \bar{\theta}]$, $|\hat{S}(\theta)| = |\sum_n a_n \mathrm{e}^{-j(n-n_c)\theta}|$, whose derivative with respect to θ is at most J in magnitude. Every θ lies within $h/2$ of a grid point, so $|\hat{S}|$ exceeds its largest grid value by at most $hJ/2$. *High frequencies.* On $[\bar{\theta}, \pi]$, let $z = \mathrm{e}^{-j\theta}$ and $Z_n = \sum_{m=1}^{n} z^m$, so that $|Z_n| = |z(1-z^n)/(1-z)| \leq 2/|1-z| = 1/\sin(\theta/2)$. Summation by parts gives $\sum_{n\geq 1} a_n z^n = \sum_{n\geq 1}(a_n - a_{n+1}) Z_n$, and $\sum_{n>N} |a_n - a_{n+1}| \leq |a_N| + 2\sum_{n>N} |a_n| \leq |a_N| + 2\tau$. So $|S - a_0| \leq B/\sin(\theta/2) \leq B/\sin(\bar{\theta}/2)$ with $a_0 = 1$. *Convergence.* For a fixed n_c, J stays bounded as $N \to \infty$, so $hJ \to 0$, and the remaining variation tends to zero. □

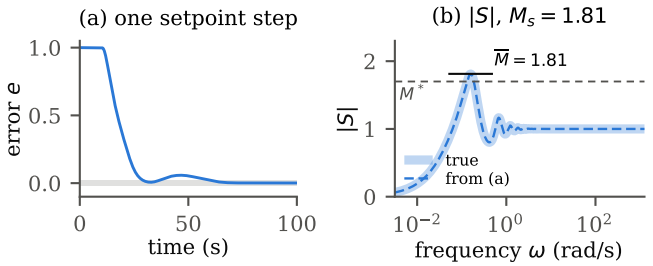


Fig. 1. One step certifies M_s, on the running example: P_3 with PID under the final tuning of basic SPIN. (a) The error after a unit setpoint step; it never crosses the setpoint (grey band: ±2%). (b) $|\hat{S}|$ from the record lies on the true $|S|$, and the bounds (2), $1.809 \leq M_s \leq 1.813$ ($\tau = 10^{-3}$), show $M_s > M^* = 1.7$: the tuning fails the test.

Everything in (2) except τ comes from the record; n_c only affects tightness (we take the weighted median of $|a_n|$). All grid values follow from one chirp z-transform, an FFT-type algorithm; the high-frequency term needs no grid at all. Fig. 1 shows the bounds on the example that runs through the paper: plant P_3 of Table II with PID, tuned by basic SPIN.

III. A Certified Acceptance Test

An iterative tuner applies a controller C, records the error after a setpoint step, and proposes the next controller from the record. We call a set of gains a *tuning*. The *test* adds one check to any tuner: a tuning *passes* if the upper bound $\overline{M}$ of Theorem 1, computed from its own record, is at most the target M^*. A tuning that fails is lowered, by the tuner's own rule or by a *uniform back-off* $C \to (1 - \beta)C$, $0 < \beta < 1$, which scales all gains down by the same factor. A tuner *accepts* a tuning when it keeps it as a candidate result; with the test, it accepts only tunings that pass.

From here on, the plant is stable (so its static gain $P(1)$ is finite), as are the self-regulating processes that SPIN targets, and the controller has integral action, $C = C_0 + K_i T_s/(z-1)$ with $K_i > 0$ and a stable proportional and derivative part C_0. So that the test can be passed, we choose $\bar{\theta}$ and the grid in (2) such that neither the high-frequency term nor the grid term can push $\overline{M}$ above M^* by itself: $1 + B/\sin(\bar{\theta}/2) = M^*$ ($\bar{\theta} = \pi$ if $1 + B > M^*$), and $hJ/2 \leq \eta$ for a small tolerance η.

Theorem 2 (Certified acceptance): Let every record meet Lemma 1 and Assumption 1. Then every tuning that passes the test has $M_s \leq M^*$, whatever rule proposed it.

Proof: Passing means $\overline{M} \leq M^*$, and $M_s \leq \overline{M}$ by Theorem 1. □

So the test never passes a fragile tuning. Nor can it trap a tuner: a uniform back-off shrinks the Nyquist curve of L towards the origin, away from -1, so it always leads to a tuning that passes.

Proposition 1 (A back-off restores the test): Let $K_i P(1) > 0$ (the integral action counteracts the error), and let every record meet Lemma 1 and Assumption 1.

(i) For a controller C, let $\mu = \min\{0, \inf_{0<\theta\le\pi} \operatorname{Re} L(\mathrm{e}^{j\theta})\}$, the leftmost real part of its Nyquist curve. For $0 < \lambda < 1/|\mu|$, the controller λC stabilizes the plant, and

$$M_s(\lambda C) \le \frac{1}{1-\lambda|\mu|}. \tag{3}$$

(ii) If $M^* > 1+2\tau+\eta$, then at most $j^* = \lceil \ln \frac{|\mu|M'}{M'-1} / \ln \frac{1}{1-\beta} \rceil$ uniform back-offs from C lead to a tuning that passes, where $M' = M^* - 2\tau - \eta$.

Proof: (i) First, μ is finite: L is bounded on the unit circle except near $z = 1$, where the integral term has the real part $-K_i T_s/2$, since $\operatorname{Re} 1/(\mathrm{e}^{j\theta} - 1) = -\frac{1}{2}$, and P is continuous at $z = 1$. For every θ, $|1+\lambda L| \ge 1+\lambda \operatorname{Re} L \ge 1-\lambda|\mu| > 0$, which gives (3). For stability, we apply the Nyquist criterion on the unit circle with a small indentation that passes outside the pole $z = 1$. On the indentation, $\lambda L \approx \lambda K_i T_s P(1)/(z-1)$ has a positive real part, since $K_i P(1) > 0$. So the whole Nyquist curve of λL lies in $\operatorname{Re} \ge -\lambda|\mu| > -1$ and cannot encircle -1; as λL has no poles outside the indented circle, the loop is stable. (ii) After j back-offs, $\lambda = (1-\beta)^j$, and (3) gives $M_s \le M'$ once $j \ge j^*$. Then the grid term of (2) is at most $(M_s + \tau) + \eta + \tau \le M^*$, and the high-frequency term is at most M^* by the choice of $\bar{\theta}$; so the tuning passes. □

The test checks a tuning only after it has been applied: the tuner learns that a tuning is fragile only by running it. But the record of C gives S of the running loop, and S determines the sensitivity of any other controller C' on the same plant. Let $\rho = C'/C - 1$ be the relative change of the controller, so that $L' = L(1+\rho)$. Then $1+L' = 1+L+L\rho = (1+L)(1+T\rho)$, and

$$S' = \frac{S}{1+T\rho}. \tag{4}$$

This is exact: it predicts S' from S and the planned change ρ, without a model of the plant. The record gives $\hat{S}$ and $\hat{T} = 1 - \hat{S}$, each within τ; Theorem 3 accounts for this error.

Theorem 3 (Certify before applying): On a stable plant, let C and C' be controllers with integral action, as in Section III, and the same T_s, let ρ be bounded on the unit circle, and let the record of C meet Lemma 1 and Assumption 1. If, for all $\theta \in (0, \pi]$,

$$d(\theta) = \min_{0\le\lambda\le1} |1 + \lambda\hat{T}\rho| - \tau|\rho| > 0, \tag{5}$$

then C' stabilizes the plant, and

$$M_s(C') \le \sup_{0<\theta\le\pi} \frac{|\hat{S}| + \tau}{|1+\hat{T}\rho| - \tau|\rho|}. \tag{6}$$

In words: (5) says that, as the controller moves gradually from C to C', the Nyquist curve never passes through -1; its minimum is the distance from 0 to the segment between 1 and $1 + \hat{T}\rho$. The bound (6) is (4) with the record error τ added.

Proof: The controllers $C_\lambda = C(1+\lambda\rho) = (1-\lambda)C + \lambda C'$ move from C to C' as λ goes from 0 to 1. Each has a positive integral gain, and its other poles are those of C_0 and C_0', inside the unit disc. So $L_\lambda = C_\lambda P$ has no poles outside the indented circle of Proposition 1, and on the indentation it has a large positive real part. By (4) with $\lambda\rho$ in place of ρ, $1 + L_\lambda = (1+L)(1+\lambda T\rho)$. Here $1 + L \ne 0$, and since $|T - \hat{T}| = |S - \hat{S}| \le \tau$, $|1 + \lambda T\rho| \ge |1 + \lambda\hat{T}\rho| - \lambda\tau|\rho| \ge d > 0$. So $1 + L_\lambda \ne 0$ on the indented circle for every λ, and the number of times its Nyquist curve encircles 0 cannot change with λ. The loop is stable at $\lambda = 0$, hence at $\lambda = 1$. The bound (6) follows from (4), $|S| \le |\hat{S}| + \tau$ and $|1 + T\rho| \ge |1 + \hat{T}\rho| - \tau|\rho|$. □

In practice, (5) and (6) are checked on the grid of Theorem 1, with margins for the gaps between grid points; above $\bar{\theta}$, $|S - 1| \le r = B/\sin(\theta/2)$ (proof of Theorem 1) gives $|T| \le r$, so (5) holds if $r|\rho| < 1$, and $|S'| \le (1+r)/(1 - r|\rho|)$. Since (4) is exact, Theorem 3 needs no small-step assumption; a larger change only widens the bound through $\tau|\rho|$. As $\tau \to 0$, (6) becomes $\max_\theta |\hat{S}/(1+\hat{T}\rho)|$, which we call the *prediction* of $M_s(C')$. A uniform back-off is the case $\rho = -\beta$.

IV. Application to SPIN

A. The SPIN method

SPIN [7] tunes an I, PI or PID controller with the integral, proportional and derivative gains K_i, K_p, K_d. In each iteration, it applies one setpoint step, records the error e_n, and forms three phase portraits, curves in the plane traced as n grows (the Pachner plots of [7]):

$$\begin{aligned} &\Gamma_0 : \big(e_n, \textstyle\sum_{m\le n} e_m\big), \qquad \Gamma_1 : (\Delta e_n, e_n), \\ &\Gamma_2 : (\Delta^2 e_n, \Delta e_n), \end{aligned} \tag{7}$$

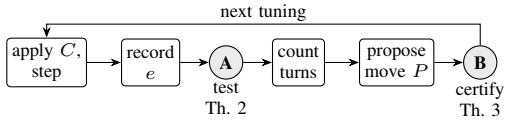
with the differences $\Delta e_n = e_n - e_{n-1} = a_n$ and $\Delta^2 e_n = \Delta e_n - \Delta e_{n-1}$. Γ_0, Γ_1 and Γ_2 emphasize low, middle and high frequencies, where the integral, proportional and derivative actions dominate the controller; we call them bands 0, 1 and 2. Each axis is divided by its peak, and the count N_k is the number of turns that Γ_k makes around its end point until the error stays within $\pm\delta$ of the setpoint (δ is 2% of the step). In short, SPIN raises all gains while every portrait turns little, and otherwise lowers the gain of the lowest band that turns too much; Algorithm 1 states the rule.[1]

B. From certified to safe SPIN

Algorithm 1 adds the tests of Section III in two steps. *Certified SPIN accepts a tuning only if it passes both the counts and the test.* It adds layer A: the counts shape the tuning, and the test decides whether it is accepted. A tuning that passes the counts but fails the test is lowered by the band move that (4) predicts to lower M_s, or else by a back-off (A2).

Corollary 1 (Certified SPIN): Under the conditions of Proposition 1, the result of certified SPIN has $M_s \le M^*$, whatever the limits $\bar{N}_k$. If no gain reaches the gain box, the test never traps SPIN: every failed tuning is followed by a tuning that passes after finitely many moves.

[1] An online simulator: `https://robopid-simulator.uceeb.cvut.cz`



Algorithm 1: one iteration (top: where the tests A and B sit): from the tuning C in use to the next tuning; **return** ends the iteration. *Basic SPIN* [7]: lines 1–7. *Certified SPIN*: adds layer A, the test (Theorem 2). *Safe SPIN*: adds A1 and layer B, which certifies each move before applying it (Theorem 3).

1: apply a setpoint step with C; record the error e
2: **if** e does not decay **then return** retreat(C)
A1: C passes the test iff $\overline{M} \le M^*$ (Theorem 1)
B1: **if** C fails **then return** the last tuning that passed (back-off(C) if none)
3: count the turns N_0, N_1, N_2
4: **if** $N_k \le \bar{N}_k$ for every band k **then**
A2: **if** C fails **then return** a band move that (4) predicts to lower M_s (back-off(C) if none)
5: $P \leftarrow$ raise(C)
6: **else** $P \leftarrow$ band move k, the lowest band with $N_k > \bar{N}_k$
7: **return** P
B2: (replaces 7) **return** the first of P, band moves $2, 1, 0$, back-off(C) that Theorem 3 certifies ($M_s \le M^*$); C if none

Actions: each multiplies the gains as listed, with $g = 1/(1-\beta)$ ($\cdot$: unchanged). Bands 0, 1, 2 belong to K_i, K_p, K_d; PI has bands 0, 1.

action	K_i	K_p	K_d	used in
raise	g	g	g	5
band move 0	$1/g$	$\cdot$	$\cdot$	6, A2, B2
band move 1	g	$1/g$	$\cdot$	6, A2, B2
band move 2	g	g	$1/g$	6, A2, B2
back-off	$1/g$	$1/g$	$1/g$	B1, A2, B2
retreat	$1/2$	$1/4$	$1/8$	2

Gains stay within $[0.001, 10]$ times their initial values (the gain box). e *does not decay* if $\max|e_n|$ over the last fifth of the record exceeds half of $\max|e_n|$. *Details:* A2 takes the highest such band, and backs off after 10 band moves without a pass; a tuning that failed B1 is not proposed again. *Accepted* tunings pass line 4 (basic), lines 4 and A1 (certified), or A1 (safe); Section V reports the most aggressive one accepted in the last 60 iterations. Defaults: $\bar{N}_k = (0.5, 0.75, 1.0)$, $\delta = 0.02$, $\beta = 0.1$, $M^* = 1.7$, $\tau = 10^{-3}$, $\eta = 2 \cdot 10^{-3}$.

Proof: The result is an accepted tuning, so it passed the test, and Theorem 2 applies. After at most 10 band moves, A2 backs off, which leads to a tuning that passes within j^* back-offs by Proposition 1(ii), unless line 2 or 6 moves the gains first. □

Certified SPIN ends safe, but it learns that a tuning is fragile only by applying it. *Safe SPIN certifies every move before applying it.* It adds layer B: it applies the test to every record and returns to the last tuning that passed when one fails (B1; A2 then never acts), so moves start only from tunings that passed; and it applies a move only if Theorem 3 certifies it (B2).

Corollary 2 (Safe SPIN): Let the plant be stable with $K_i P(1) > 0$, let the first tuning that passes the test stabilize it, and let the record of every stable loop meet Assumption 1. Then, from that tuning on, every tuning that safe SPIN applies stabilizes the plant and has $M_s \le M^*$, whatever the limits $\bar{N}_k$.

Proof: By induction over the iterations. Let the tuning C in use be stable; by B1, either C passes, so $M_s(C) \le M^*$ (Theorem 2), or the next tuning is the last one that passed, stable with $M_s \le M^*$. If C passes, B2 applies either C itself or a move that Theorem 3, from the record of C, certifies to be stable with $M_s \le M^*$. □

C. Why the turn counts alone cannot certify

The counts see setpoint crossings, and crossings reveal M_s only on special loops. To show this, write a portrait (7) as (p, q), with q the running sum of p, shifted so that the end point is the origin. The count is the net counterclockwise angle swept around the origin, divided by 2π. We state the results in continuous time, where $\dot{q} = p$. For Γ_0, $p = e$, and its zeros are the setpoint crossings.

Lemma 2 (Half turns): Let $\dot{q} = p$ with p continuous, let $p(t_1) = p(t_2) = 0$ with $q(t_1)q(t_2) < 0$, and let $p \neq 0$ on (t_1, t_2). Then, on $[t_1, t_2]$, the curve (p, q) turns counterclockwise around the origin by exactly half a turn, and so does its image under any linear map with a positive determinant, such as the per-axis scaling of SPIN.

Proof: On (t_1, t_2), p keeps one sign, so the curve stays in one half-plane bounded by the q-axis, and q moves monotonically from one ray of the q-axis to the opposite ray. The curve avoids the origin, since $p = 0$ only at t_1 and t_2, where $q \neq 0$. The angle thus changes by exactly π, counterclockwise, since q increases where $p > 0$ and decreases where $p < 0$. A linear map with a positive determinant maps half-planes to half-planes and opposite rays to opposite rays, and keeps the orientation. □

So each setpoint crossing adds half a turn to N_0, and a count sees the crossings that rise above δ. Whether crossings reveal M_s depends on the loop. We first study a loop where they do. Lambda, IMC and SIMC tuning rules cancel the plant lag by the controller zero [2], e.g., PI with the integral time $T_i = T_1$ on the plant $K\mathrm{e}^{-sL_d}/(T_1 s + 1)$. The loop is then

$$L(s) = \frac{\kappa\, \mathrm{e}^{-sL_d}}{sL_d}, \qquad \kappa = \frac{K K_p L_d}{T_i}. \tag{8}$$

Proposition 2 (Crossings on lag-cancelled loops): The loop (8) is stable for $0 < \kappa < \pi/2$. Its M_s strictly increases with κ and is given by

$$\kappa = x \sin x - x^2 \cos x, \qquad \frac{1}{M_s^2} = 1 - \frac{2\kappa \sin x}{x} + \frac{\kappa^2}{x^2}, \tag{9}$$

where $x \in (0, \pi/2)$ is the peak frequency normalized as $x = \omega L_d$. The error after a unit setpoint step never crosses the setpoint if and only if $\kappa \le 1/\mathrm{e}$, that is, $M_s \le 1.394$; otherwise it crosses infinitely often.

Proof: At $x = \omega L_d$, L has the phase $-\pi/2 - x$ and the gain κ/x. The phase first reaches $-\pi$ at $x = \pi/2$, where the gain is $2\kappa/\pi < 1$, so the loop is stable. Let $f(x) = |1 + L|^2 = 1 - 2\kappa \sin x/x + \kappa^2/x^2$, so that $M_s^2 = 1/\min_x f$. Then $f'(x) = 2\kappa(\varphi(x) - \kappa)/x^3$ with $\varphi(x) = x \sin x - x^2 \cos x$,

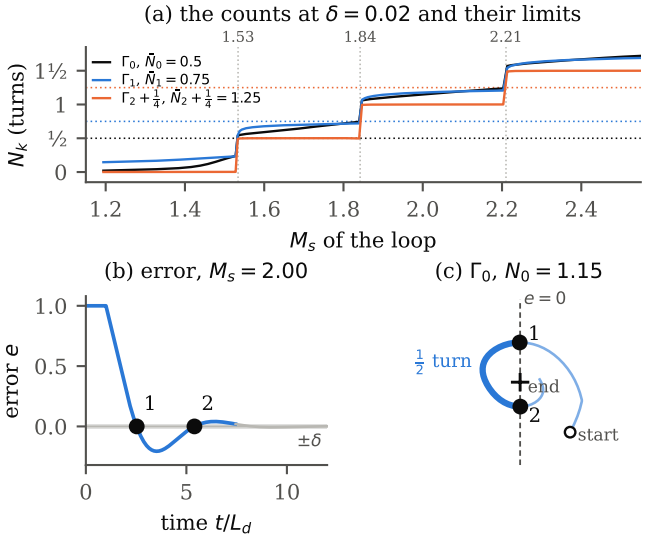


Fig. 2. SPIN's counts on the lag-cancelled loop (8). (a) Each count steps over its limit (dotted) at $M_s = 1.53$, 1.84, 2.21 (N_2 and its limit shifted up a quarter turn). (b) A record with $M_s = 2.00$ and two crossings. (c) Its portrait Γ_0 turns by exactly half a turn between them (Lemma 2); the radius is compressed for display.

which increases on $(0, \pi)$ and has $\varphi(\pi/2) = \pi/2 > \kappa$. So f has one minimum on $(0, \pi)$, at the $x^* \in (0, \pi/2)$ with $\varphi(x^*) = \kappa$. Elsewhere, f is larger: $f \geq 1$ on $[\pi, 2\pi]$, where $\sin x \leq 0$, and for $x \geq 2\pi$, $f \geq (1-\kappa/x)^2 > (1-2\kappa/\pi)^2 = f(\pi/2)$. This gives (9). As κ grows, $f(x^*)$ falls, since its derivative is $\partial f/\partial\kappa = -2\cos x^* < 0$ (envelope theorem). For the crossings, the output obeys $\dot{y}(t) = (\kappa/L_d)\, e(t-L_d)$, so the error obeys $\dot{e}(t) = -(\kappa/L_d)\, e(t-L_d)$ with $e = 1$ on $[0, L_d]$. It stays positive if and only if its characteristic equation $s+(\kappa/L_d)\mathrm{e}^{-sL_d} = 0$ has a real root; with $w = sL_d$, this reads $-w\mathrm{e}^w = \kappa$, and $-w\mathrm{e}^w$ has the maximum $1/\mathrm{e}$ over real w. Otherwise all solutions oscillate [17]. □

So, at $\delta = 0.02$, the counts of the loop (8) rise in steps as M_s grows (Fig. 2(a)): with n crossings larger than δ, N_0 and N_1 lie within a quarter turn above $n/2$, and N_2 is $n/2-1/4$ within 0.01 (Γ_2 starts with a quarter turn clockwise, as $\dot{e}$ jumps at $t = L_d$). The limits $\bar{N}_0 = 1/2$, $\bar{N}_1 = 3/4$, $\bar{N}_2 = 1$ of [7] allow 0, 1 and 2 crossings, that is, $M_s \leq 1.534$, 1.842 and 2.210; Γ_0 binds first, so on these loops SPIN stops at $M_s \leq 1.534$.

On general loops, this guarantee is lost. The counts leave the steps (in the 3800 count decisions of certified SPIN in Section V, N_0 and N_2 lay off them in 23 and 69%), and crossings do not bound M_s: the basic tuning of Fig. 1 never crosses the setpoint, yet has $M_s = 1.81$ (a PI loop on a first-order plant reaches $M_s = 2.62$ without a crossing above 2%), while a controller zero can cause crossings at a low M_s. The counts can shape a tuning, but only a certificate can certify it.

V. Simulations

Setup: We test the three variants on the 19 loops of Table II. Each runs 200 iterations of Algorithm 1 with its

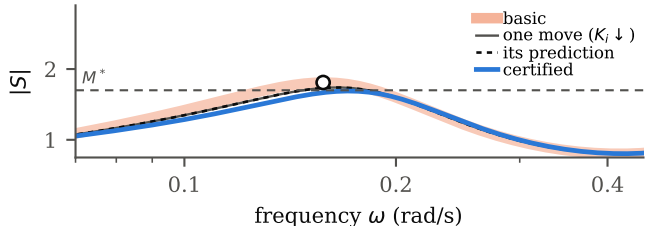


Fig. 3. $|S|$ near its peak, P_3 with PID: the basic tuning ($\overline{M} = 1.81$, ○), one band move lowering K_i ($M_s = 1.740$, as predicted by (4) from the record of the basic tuning), and the certified tuning ($M_s = 1.690$).

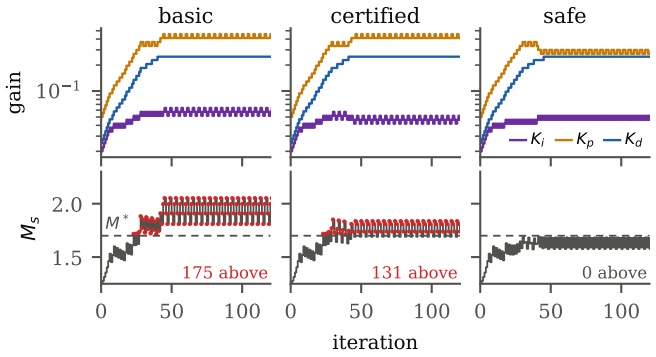


Fig. 4. Tuning history, P_3 with PID, cautious start, as the online simulator shows it: the gains (top) and M_s of the tuning applied in each iteration (bottom; dots: $M_s > M^*$).

defaults. The gain box ties the result to the start, so every loop runs from two starts: SIMC [2] on a half-rule first-order-plus-dead-time model, with the closed-loop time constant three times the model dead time (cautious), and the same tuning with all gains doubled (bold). P, I and D act on the error. As in a real controller, the sampling period stays fixed, and each record covers 15 times the dead time plus the lag time constants. M_s is that of this digital loop, computed from its exact L. Performance is measured by the load-step IAE, the integral of $|y|$ after a unit step disturbance at the plant input.

The certificates held and were tight: For all 276 accepted tunings, a four times longer record showed a tail below $3 \cdot 10^{-5}$, well within $\tau = 10^{-3}$, and every M_s lay between its bounds, at most 0.003 below $\overline{M}$ (Theorem 1). Of 456 random moves from accepted tunings, with gains scaled by up to 8, all 339 certified moves were stable and all 117 rejected ones unstable, and the bound (6) exceeded M_s by a median of 0.011 (Theorem 3).

Certified SPIN ends below M^*, only safe SPIN stays below it: Table III shows both corollaries. Basic SPIN ended above M^* on 9 and 7 of the 19 loops; certified SPIN on none, and every loop kept accepting tunings (Corollary 1). But certified SPIN must apply a tuning to reject it, so it still applied tunings above M^* in 45% of its iterations, against 57% for basic SPIN. Safe SPIN applied none, and all its tunings were stable (Corollary 2; every start tuning passed the test, so the corollary covered every iteration).

Robustness costs little with safe SPIN: Figs. 3–5 follow P_3 with PID. From the record of the basic tuning, (4)

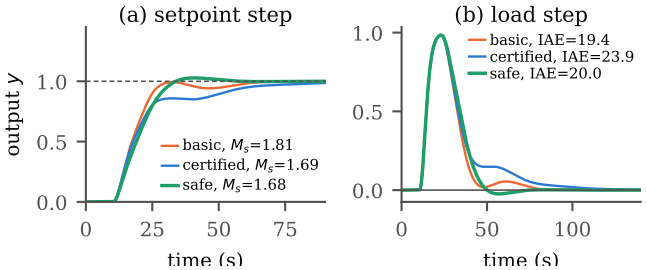


Fig. 5. The tuned loops of Fig. 4 (final tunings): output after (a) a unit setpoint step and (b) a unit load step at the plant input.

TABLE II

THE 19 TEST LOOPS

	plant $P(s)$	controller	loops
from [7]	$P_1 = \mathrm{e}^{-s}/((10s+1)(s+1)^3)$	PI, PID	2
	$P_2 = 1.25\,\mathrm{e}^{-8s}/(5s+1)^4$	PI, PID	2
	$P_3 = \mathrm{e}^{-10s}/((2s+1)(s+1)^3)$	PI, PID	2
	$P_4 = \mathrm{e}^{-4s}/(8s+1)^6$	PI, PID	2
one band	$\mathrm{e}^{-5s}/(s+1)$	PI	1
	$\mathrm{e}^{-s}/(10s+1)$	PI	1
	$\mathrm{e}^{-0.5s}/((10s+1)(s+1))$	PI	1
	$\mathrm{e}^{-0.1s}/((s+1)(0.5s+1))$ (fast)	PID	1
family	$\mathrm{e}^{-L_d s}/((s+1)(0.5s+1)(0.25s+1))$, $L_d = 0.25, 0.5, 1, 2, 4, 8, 16$	PID	7
total			19

Fixed sampling period: $T_a/1000$ for PI; for PID, the smaller of $T_a/100$ and $T_f/20$ of the start tuning, where T_a is the dead time plus the lag time constants and $T_f = T_d/10$, $T_d = K_d/K_p$, the derivative filter time constant.

predicted the M_s after a band move exactly (Fig. 3). Basic and certified SPIN climb alike (Fig. 4); certified SPIN then lowers K_i, but it must apply each tuning of its final cycle to reject it, so 131 of its 200 applied tunings exceed M^*. Safe SPIN applies only certified moves and trades K_p for K_d. The setpoint responses are alike (Fig. 5), and the basic tuning does not overshoot although its M_s is 1.81: the response alone does not reveal fragility. In load rejection, certified SPIN, with its lower K_i, is 23% worse than basic SPIN, and safe SPIN only 3%. Over all loops, safe and certified SPIN have the same median load-step IAE; the mean of safe SPIN is higher only because of the fast PID loop, where it settled at $M_s \approx 1.5$.

Count limits do not replace the certificate: With every $\bar{N}_k = 1/2$, certified SPIN still applied tunings above M^* in 1310 and 1420 iterations, accepted no tuning on 2 and 3 loops, and raised the median load-step IAE by 11 and 19%.

VI. CONCLUSION

The setpoint step that a model-free tuner applies anyway certifies the robustness of the loop: its record bounds M_s from both sides, without a model or a dedicated experiment, and certifies the stability and M_s of the next tuning before it is applied. Checking a tuning after applying it makes the result safe; checking it before makes the whole tuning safe. The theorems assume a linear loop and a long enough record; noise and real loops are left to a follow-up. Scripts: `https://github.com/ottapav/robopid-simulator`.

TABLE III

RESULTS ON THE 19 LOOPS, FROM THE CAUTIOUS (C) AND BOLD (B) START

	basic		certified		safe	
	c	b	c	b	c	b
final tuning[a]						
loops with $M_s > M^*$	9	7	0	0	0	0
largest M_s	2.01	2.07	1.69	1.69	1.69	1.69
load-step IAE[b]	–	–	+9%	+8%	+15%	+22%
while tuning (3800 iterations)						
iterations with $M_s > M^*$	2146	2225	1623	1775	0	0
largest M_s	2.35	2.38	1.93	1.98	1.696	1.695

[a]The most aggressive tuning accepted in the last 60 iterations. [b]Mean change from basic SPIN.

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